

Notes: Goldin (AER, 2014)

A Grand Gender Convergence: Its Last Chapter

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1 Big picture

- **Question.** What must happen for the remaining gender gap in labor-market pay to close? The paper argues that the answer is not mainly more schooling, stronger bargaining, or a generic anti-discrimination story. The crucial margin is how jobs are *structured* and *remunerated*.
- **Setting.** The paper is a broad empirical essay on the United States. It combines: (i) Census and ACS evidence on gender earnings gaps over the life cycle; (ii) occupation-level evidence on residual pay gaps; (iii) O*Net measures of occupational structure; and (iv) detailed professional evidence for MBAs, JDs, and pharmacists.
- **Core challenge.** The “explained” share of the gender gap has shrunk as women caught up and often surpassed men in education and labor-market preparation. But a sizable *residual* gap remains, and that remaining gap widens sharply with age and differs greatly across occupations.
- **Main conceptual contribution.** Goldin interprets the remaining pay gap through **personnel economics** and **compensating differentials**. The central idea is that some jobs reward long, continuous, and particular hours in a *nonlinear* (convex) way, while other jobs pay more nearly linearly with respect to hours.
- **Main hypotheses.**
 - When workers are poor substitutes for each other, firms have reason to reward long and inflexible hours disproportionately.
 - Workers who value temporal flexibility therefore face larger pay penalties in such occupations.
 - Because women, especially mothers, more often demand schedule flexibility or shorter hours, nonlinear pay structures generate larger gender gaps even among highly educated workers.
- **Main empirical design.** The paper is not a single-regression study. Instead, it builds the argument in layers:
 - synthetic-cohort earnings profiles over the life cycle;
 - within-occupation residual pay-gap regressions;
 - occupation-level hours elasticities and O*Net characteristics;
 - detailed profession-specific evidence from MBAs, lawyers, and pharmacists.
- **Main baseline result.** Men and women start their careers with earnings that are relatively close, but the gap widens strongly with age. This widening remains even after controlling for hours, weeks worked, and education.
- **Key heterogeneity result.** The residual gap is large in business and law, much smaller in technology and science, and especially small in pharmacy.
- **Key mechanism result.** Occupations with a stronger premium to long hours and more time pressure, interpersonal dependence, and low substitutability are precisely the occupations with the larger gender earnings gaps.

- **Main credibility result.** The same pattern shows up within narrow professional groups with common training, and the technology result is not mainly an industry-composition or female-selection artifact.
- **Economic takeaway.** The “last chapter” of gender convergence requires a labor market in which workers can substitute for each other more easily and in which earnings depend less on working *particular* hours or *very long* hours.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- The paper is useful precisely because it studies a setting where traditional human-capital differences have narrowed a lot.
- That makes the remaining pay gap more informative about *job design*, *hours premiums*, and *temporal flexibility*.
- Goldin also focuses heavily on college graduates and elite professions, where the premium to long and inflexible hours appears to be strongest.

Interpretation. The paper is asking: once men and women look much more similar in education and training, why does pay still diverge so much later in life?

2.2 Data sources and sample construction

- **US Census / ACS.** The aggregate evidence uses US Census microdata for 1970, 1980, 1990, and 2000, plus ACS microdata for 2005 and 2010.
- **Main aggregate samples.**
 - white, native-born, non-military workers;
 - typically ages 25–69 for the life-cycle figures and 25–64 for occupation regressions;
 - in some exercises full-time, full-year workers only;
 - in other exercises all workers with positive earnings and positive hours.
- **O*Net.** Used to measure occupational features such as time pressure, contact with others, discretion, and work structure.
- **National Survey of College Graduates, 2003.** Used to examine whether women with technology degrees have unusually low labor-force participation.
- **University of Chicago Booth MBA survey.** Covers MBA graduates from 1990 to 2006 and provides retrospective person-year information on earnings, hours, employment interruptions, marriage, and children.

- **University of Michigan Law School Alumni Survey.** A longitudinal sample for JD graduates from 1982 to 1991, observed at roughly years 5 and 15 after graduation.
- **National Pharmacist Workforce Surveys.** Used mainly for 2000, 2004, and 2009 to study a profession with relatively linear pay.

Interpretation. The paper deliberately combines broad labor-market data with narrow, high-detail professional data. The first shows the pattern; the second helps explain the mechanism.

2.3 Main outcome: the gender earnings gap

The main object throughout the essay is the relative earnings of women to men, often summarized as

$$G = \log \left(\frac{\bar{Y}^F}{\bar{Y}^M} \right). \quad (1)$$

A value of $G = 0$ means equality, while more negative values mean lower female earnings relative to male earnings.

The paper examines this gap using:

- annual earnings in the aggregate data;
- annual earnings and hourly fees in the professional data;
- both raw gaps and residual gaps after adding controls.

Interpretation. The paper is interested less in whether women choose different occupations in the first place and more in why earnings differ *within* occupations and *within* homogeneous degree groups.

2.4 Synthetic cohort life-cycle profiles

- Figure 1 traces the female/male earnings ratio by age for synthetic birth cohorts of college graduates.
- Part A uses full-time, full-year annual earnings with no extra controls.
- Part B adds controls for education beyond college, log hours, and log weeks.

Interpretation. This is the paper's first major fact: the gender earnings gap is much smaller at labor-market entry than later in the life cycle.

2.5 Residual occupation-level pay gaps

To study within-occupation differences, the paper estimates log earnings regressions with:

- a female indicator,
- age as a quartic,

- race and year controls,
- log hours and log weeks,
- education controls,
- occupation dummies,
- and occupation $\times$ female interactions.

These regressions are used to recover how much of the gender gap remains within each occupation.

2.6 Occupational flexibility and substitutability measures

The O*Net exercise centers on five normalized job characteristics:

1. time pressure,
2. contact with others,
3. establishing and maintaining interpersonal relationships,
4. structured versus unstructured work,
5. freedom to make decisions.

Interpretation. These characteristics proxy for how difficult it is to replace one worker with another and how costly it is for a worker to be absent at a particular time.

2.7 Descriptive facts

- For full-time, full-year workers ages 25–69, the female/male annual earnings ratio in 2010 is about **0.72** for means and **0.77** for medians.
- For college graduates, the 2010 ratio is lower: about **0.65** for means and **0.72** for medians.
- Among college graduates working full-time, full-year, the aggregate gender gap is much smaller at career start than later in life.
- Among college graduates, roughly **58% to 68%** of the aggregate earnings gap comes from *within-occupation* differences rather than between-occupation sorting.
- For business occupations, the residual gap is much larger than for science and technology occupations.
- Pharmacy stands out as highly egalitarian: the unadjusted full-time, full-year female/male earnings ratio is about **0.85**, and the hours-adjusted ratio is around **0.93–0.95**.

Interpretation. The raw facts already point away from a simple “women choose the wrong occupations” story and toward a “how occupations pay for time” story.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper’s mechanism has two layers:

- **Layer 1: convergence in human capital.** Women have largely caught up in schooling, professional training, and labor-market experience.
- **Layer 2: differential pricing of flexibility.** Some jobs make flexibility expensive because output depends on long, continuous, and synchronized hours.

Thus the modern residual gap is interpreted less as a pure skill gap and more as the market price of temporal flexibility.

3.2 Residual earnings regressions in the aggregate data

A generic version of the paper’s occupation-level regression can be written as

$$\log Y_{iot} = \alpha + \beta Female_i + f(Age_i) + X_i' \gamma + \theta \log Hours_i + \lambda \log Weeks_i + \sum_o \delta_o \mathbf{1}\{Occ_i = o\} + \varepsilon_{iot}, \quad (2)$$

where Y_{iot} is annual earnings, X_i includes race, year, and education controls, and the occupation dummies absorb broad occupational sorting.

To allow gender gaps to differ across occupations, the paper adds

$$\log Y_{iot} = \dots + \sum_o \phi_o \left(Female_i \times \mathbf{1}\{Occ_i = o\} \right) + \varepsilon_{iot}. \quad (3)$$

Interpretation.

- β captures the residual female penalty in the reference group.
- ϕ_o captures how that penalty differs in occupation o .
- Comparing regressions with and without occupation dummies tells us how much of the aggregate gap is between occupations versus within occupations.

3.3 Occupation-specific elasticity of earnings with respect to hours

To measure whether earnings are convex in hours, Goldin effectively allows the return to hours to differ by occupation:

$$\log Y_{iot} = \dots + \sum_o \kappa_o \left(\log Hours_i \times \mathbf{1}\{Occ_i = o\} \right) + \varepsilon_{iot}. \quad (4)$$

Here κ_o is the elasticity of annual earnings with respect to weekly hours in occupation o .

Interpretation.

- If $\kappa_o \approx 1$, earnings are roughly linear in hours.
- If $\kappa_o > 1$, longer hours receive a disproportionate premium, meaning the earnings schedule is convex.
- Convexity is precisely what makes modest reductions in hours translate into large earnings penalties.

3.4 Personnel-economics framework: why convexity emerges

The paper's core theoretical object is output in occupation j for worker i :

$$Q = \begin{cases} \lambda_i k_j, & \text{if } \lambda_i > \lambda_j^*, \\ \lambda_i k_j (1 - \delta_j), & \text{if } \lambda_i \leq \lambda_j^*, \end{cases} \quad (5)$$

where:

- λ_i is the worker's time input or temporal availability,
- k_j is output per unit of time in occupation j ,
- λ_j^* is the threshold below which lower availability becomes especially costly,
- δ_j is the productivity loss from insufficient continuity or timing.

Goldin then considers three positions: a high-slope, high-penalty job; a middle job; and a reservation job with linear pay ($\delta_r = 0$). Workers who want lower or more flexible hours sort into positions with lower slopes but also lower penalties.

Interpretation.

- The most lucrative jobs are often exactly the ones with the highest flexibility penalty.
- The model therefore predicts both occupational sorting and convex within-occupation pay schedules.

3.5 O*Net mapping from job structure to the gender gap

The paper also relates the residual occupation-level gender gap to a summary index of O*Net characteristics:

$$Gap_o = a + b ONET_o + u_o. \quad (6)$$

In Figure 5, the fitted line is approximately

$$Gap_o = -0.153 - 0.0593 ONET_o, \quad (7)$$

with $R^2 \approx 0.214$.

Interpretation. Occupations that are more time-pressured, relational, structured, and discretion-intensive have more negative female relative earnings.

3.6 Professional-sample earnings specifications

The professional evidence refines the same idea within homogeneous groups.

A generic specification for the JD and MBA evidence is:

$$\log Y_{it} = \alpha + \beta Female_i + \theta \log Hours_{it} + \lambda \log Weeks_{it} + H'_{it}\rho + Z'_i\pi + \varepsilon_{it}, \quad (8)$$

where H_{it} includes employment interruptions, part-time history, and job tenure, and Z_i contains degree-specific performance controls.

Interpretation.

- If the female coefficient shrinks sharply once hours and interruptions are added, then the gap is largely generated by nonlinear pay and career interruptions rather than by lower contemporaneous skill alone.
- This is exactly what happens for MBAs and JDs.

3.7 Linear versus nonlinear occupations

The paper's occupational taxonomy is simple but powerful:

- **Nonlinear occupations:** MBAs, many business jobs, law.
- **Closer to linear occupations:** pharmacy, much of technology, much of science.

Interpretation. The paper's strongest claim is comparative: the gender gap is large where pay is convex and small where pay is linear.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the main claims

The paper does *not* use an instrument in the standard IV sense. Instead, credibility comes from a pattern of comparisons:

- comparisons across ages within synthetic cohorts,
- comparisons across occupations after rich controls,
- comparisons across occupations with different hours elasticities,
- comparisons within very homogeneous professional groups,

- comparisons between occupations that differ in flexibility costs.

Interpretation. The paper identifies a strong and coherent regularity rather than a single causal parameter from one design.

4.2 Why a pure human-capital explanation is incomplete

A simple human-capital story struggles with four facts:

- men and women begin much closer in pay than they end up,
- the gap widens after the childbearing years begin,
- women without children earn much more similarly to comparable men,
- occupations with the same broad educational level have dramatically different residual gaps.

Interpretation. Human capital still matters, but it no longer seems to be the main explanation for the remaining high-skill gender gap.

4.3 Within-occupation versus between-occupation evidence

Occupation sorting does matter, but Table 1 shows it is not the main story.

- Adding a full set of occupation dummies reduces the female coefficient by only about one-third or less.
- Among college graduates, most of the earnings gap is due to *within-occupation* differences.

Interpretation. Moving women into “male” occupations is not sufficient if those occupations continue to reward inflexible time disproportionately.

4.4 Selection concerns in technology occupations

A natural objection is that technology appears egalitarian only because women who remain there are unusually selected.

Goldin addresses this using the National Survey of College Graduates:

- women with technology degrees do *not* have lower labor-force participation than other college-graduate women,
- if anything, their participation is slightly higher,
- the arrival of young children reduces participation less for women with technology degrees.

Interpretation. Selection does not appear to be the dominant reason technology occupations look more egalitarian.

4.5 Occupation versus industry in technology

Another concern is that technology workers may simply be employed in especially progressive or high-productivity industries.

The paper shows:

- technology workers in technology-heavy industries earn more,
- but women do not earn disproportionately more within those industries,
- the greater equality is linked more to *technology occupations* than to *technology industries*.

Interpretation. The key unit is the job, not merely the sector.

4.6 Why the professional evidence is powerful

The MBA and JD samples dramatically reduce heterogeneity in training.

- The workers have the same advanced degree.
- In law, they come from the same law school and the regressions can condition on law-school GPA and law review.
- The surveys contain detailed histories of part-time work, nonemployment, hours, and children.

Interpretation. Once hours and interruption histories are measured carefully, much of the late-career gender gap in these professions becomes understandable.

4.7 What the paper can and cannot distinguish

- The paper does *not* claim that discrimination is irrelevant.
- It does *not* fully separate preferences from social norms inside households.
- It does *not* estimate one universal structural parameter for the cost of flexibility.

What it *does* establish is that the modern high-skill gender gap is strongly organized around temporal flexibility, hours convexity, and worker substitutability.

4.8 Relation to the literature

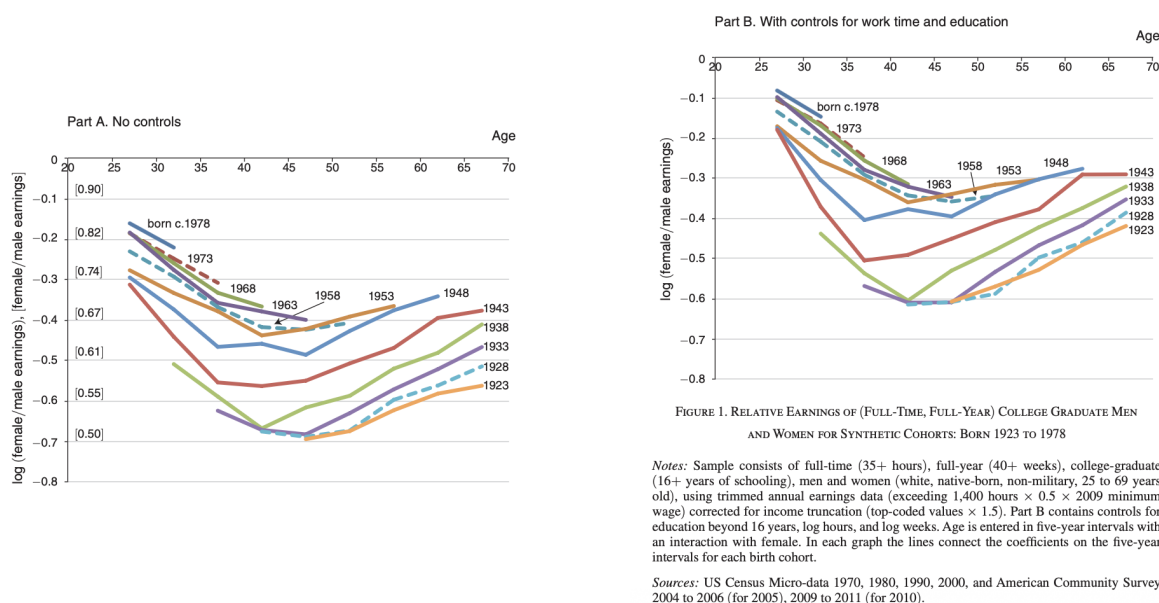
The paper links:

- classic gender-wage decompositions,
- human-capital and family-interruption models,
- personnel economics,
- and compensating differentials.

Its distinctive contribution is to make the *structure of work*—not just worker characteristics—the centerpiece of the explanation.

5 Tables and figures

5.1 Figure 1: relative earnings over the life cycle for synthetic cohorts



Read:

- Each younger cohort begins with a higher female/male earnings ratio than the cohort before it.
- For recent college-graduate cohorts, women begin with earnings close to men, especially once controls for hours, weeks, and education are added.
- The ratio then falls sharply through the thirties and forties.
- Even with controls, the life-cycle divergence remains substantial.

Economic message. The central modern gender-gap fact is not that women start far behind. It is that the gap opens up *after* labor-market entry.

5.2 Table 1: residual gender differences in earnings and the role of occupation

Read:

- For full-time college graduates, the raw female coefficient is about -0.285 .

TABLE 1—RESIDUAL GENDER DIFFERENCES IN EARNINGS AND THE ROLE OF OCCUPATION

Sample	Variables included	Coefficient on female	Standard error	R ²
Full-time	Basic	−0.248	0.00101	0.112
Full-time	Basic, time	−0.193	0.00100	0.163
Full-time	Basic, time, education	−0.247	0.000905	0.339
Full-time	Basic, time, education, occupation	−0.192	0.00104	0.453
All	Basic	−0.320	0.00105	0.102
All	Basic, time	−0.196	0.000925	0.353
All	Basic, time, education	−0.245	0.000847	0.475
All	Basic, time, education, occupation	−0.191	0.000963	0.563
Full-time, BA	Basic	−0.285	0.00159	0.131
Full-time, BA	Basic, time	−0.230	0.00158	0.177
Full-time, BA	Basic, time, education	−0.233	0.00155	0.216
Full-time, BA	Basic, time, education, occupation	−0.163	0.00158	0.374
All, BA	Basic	−0.384	0.00173	0.119
All, BA	Basic, time	−0.227	0.00151	0.380
All, BA	Basic, time, education	−0.229	0.00148	0.407
All, BA	Basic, time, education, occupation	−0.163	0.00151	0.525

Notes: "Basic" regression is the log of annual earnings regressed on the female dummy, age as a quartic, race, and year. "Time" adds log hours per week and log weeks. "Education" adds dummies for education categories (and those above a BA for the college graduate sample). "Occupation" adds three-digit occupation dummies. "Full-time" is 35 and above hours per week and 40 and above weeks per year. The "All" includes workers 25 to 64 years old with positive earnings and positive hours worked during the past year. The "full-time" sample consists of full-time, full-year individuals 25 to 64 years old excluding those in the military using trimmed annual earnings data (exceeding 1,400 hours $\times$ 0.5 $\times$ 2009 minimum wage). The "BA" sample includes workers with at least a college or university bachelor's degree. The number of observations is 2,603,968 for full-time, 3,291,168 for all, 964,705 for full-time BA or more, and 1,162,638 for all BA or more.

Source: American Community Survey 2009 to 2011.

- Adding log hours and log weeks reduces it to about -0.230 .
- Adding education beyond the BA changes little.
- Adding occupation dummies reduces it further to about -0.163 .
- For all college graduates (not just full-time), the coefficient falls from about -0.384 to -0.163 once time, education, and occupation controls are included.

Economic message. Occupation matters, but most of the problem remains *within* occupations rather than across them.

5.3 Figure 2: gender pay gaps by occupation

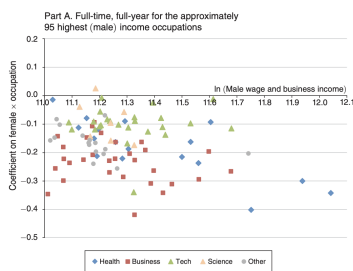


FIGURE 2A. GENDER PAY GAPS BY OCCUPATION: 2009 TO 2011

Notes: Sample consists of full-time, full-year individuals 25 to 64 years old excluding those in the military using trimmed annual earnings data (exceeding 1,400 hours $\times$ 0.5 $\times$ 2009 minimum wage). Regression contains age in a quartic, race, log hours, log weeks, education levels, census year, all occupations (469), and an interaction with female and occupation. Part A contains all full-time, full-year workers (2,603,968 observations); part B has those who graduated (BA) college (964,705 observations); part C has the group < 45 years old among those included in part A (1,333,013 observations). Each of the symbols in part A is an occupation for which the mean annual income for males exceeds \$60K (current \$) and is limited to occupations with at least 25 males and at least 25 females. For parts B and C the same occupations are graphed.

Source: American Community Survey 2009 to 2011.

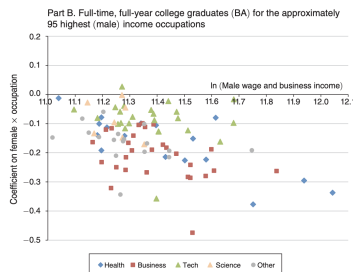


FIGURE 2B

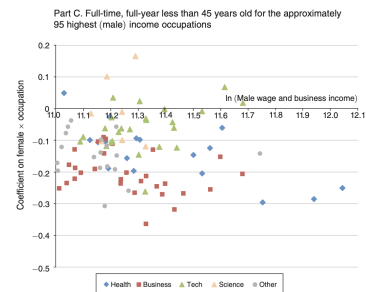


FIGURE 2C

Read:

- Almost all occupation-specific coefficients on female are negative.
- Business occupations have the largest negative residual coefficients.

- Technology and science occupations have the smallest residual gaps.
- Among workers younger than 45, some science and technology occupations even have coefficients near zero or above zero.

Economic message. Occupations differ enormously in how heavily they penalize being a worker who needs flexibility.

5.4 Figure 3 and Table 2: convexity, O*Net characteristics, and the size of the gap

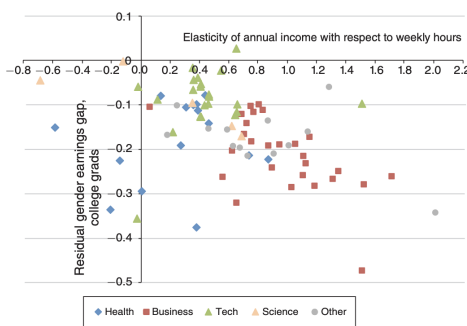


FIGURE 3. RELATIONSHIP BETWEEN THE ELASTICITY OF EARNINGS WITH RESPECT TO HOURS AND THE GENDER EARNINGS GAP

Notes: The controls are as follows: Age as a quartic, race, $\ln(\text{hours})$, $\ln(\text{weeks})$, education, ACS years, occupation and occupation $\times$ female; 35+ hours, 40+ weeks; 25 to 64 year old college graduates. Only occupations with more than 25 males and more than 25 females are graphed. The vertical axis is the coefficient on occupation $\times$ female from Figure 2, part B. The computation of the elasticity of annual income with respect to weekly hours by occupation adds to the regression in Figure 2, part B the interaction of $\ln(\text{hours})$ with occupation. The regressions include all three-digit occupations but just the categorized group of about 95 occupations is shown.

Source: American Community Survey 2009 to 2011 and Figure 2, part B.

TABLE 2—O*NET CHARACTERISTICS: MEANS (NORMALIZED) BY OCCUPATIONAL GROUP

O*Net characteristics	Technology and science	Business	Health	Law
1. Time pressure	−0.488	0.255	0.107	1.51
2. Contact with others	−0.844	0.171	0.671	0.483
3. Establishing and maintaining interpersonal relationships	−0.611	0.548	0.276	0.781
4. Structured vs. unstructured work	−0.517	0.313	0.394	1.22
5. Freedom to make decisions	−0.463	−0.00533	0.974	0.764
Number of occupations	31	28	16	1

Notes: The occupations are those in Figure 2, part A. When there is more than one O*Net occupation for an ACS occupation, the characteristic is weighted by the number of workers in each of the O*Net occupations. Each of the O*Net characteristics has been normalized to have a mean of 0 and a standard deviation of 1. The work setting characteristics and questions most relevant to the issues raised here are:

1. Time pressure: *How often does this job require the worker to meet strict deadlines?*
Lower pressure means worker does not have to be around at particular times.
2. Contact with others: *How much does this job require the worker to be in contact with others (face-to-face, by telephone, or otherwise) in order to perform it?*
Less contact means greater flexibility.
3. Establishing and maintaining interpersonal relationships: *Developing constructive and cooperative working relationships with others, and maintaining them over time.*
The more working relationships, the more workers and clients the employee must be around.
4. Structured versus unstructured work: *To what extent is this job structured for the worker, rather than allowing the worker to determine tasks, priorities, and goals?* If the job is highly structured to the worker, there would be a lower chance that the worker would have close substitutes.
5. Freedom to make decisions: *How much decision making freedom, without supervision, does the job offer?*
Generally means that the worker determines what each client should receive, rather than being given a specific project, and thus workers are poorer substitutes for each other the greater are these freedoms.

Source: O*Net Online, <http://www.onetonline.org/>

Read:

- Figure 3 shows a negative relationship between the residual gender gap and the elasticity of earnings with respect to hours.
- Occupations with elasticities above one—especially in business and law—have larger negative gender gaps.
- Table 2 shows that technology and science occupations score much lower than business, health, and law on time pressure, contact with others, interpersonal dependence, work structure, and decision freedom.
- Figure 5 summarizes the O*Net relationship: occupations with more of these features have worse female relative pay.

Economic message. The gap is large where jobs are hard to hand off, teamwork is costly to interrupt, and long hours receive superlinear rewards.

5.5 Figure 4: a theory of occupational pay differences

Read:

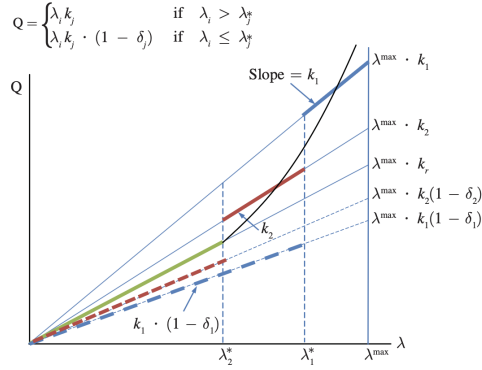


FIGURE 4. A THEORY OF OCCUPATIONAL PAY DIFFERENCES

Notes: Each of the lines gives the relationship between output, Q , in some occupation and the time input, λ , of a worker where $0 < \lambda \leq \lambda^{\max}$. When the time input is reduced below some level, λ^* , output decreases discretely for occupations 1 and 2. Occupation r has a linear relationship between time worked and earnings throughout. An individual who works between λ_1^* and $\lambda^{\max}$ will be in occupation 1, an individual between λ_1^* and λ_2^* will work in 2 and all others will work in r , if they remain in the labor force.

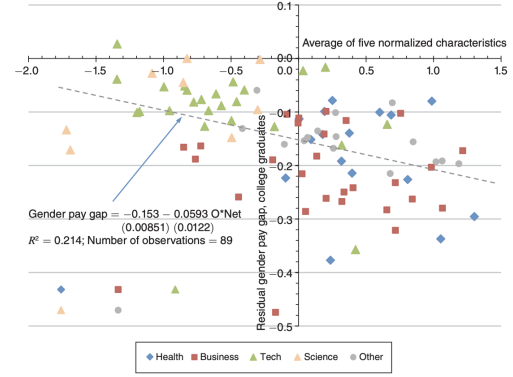


FIGURE 5. O*Net CHARACTERISTICS AND THE RESIDUAL COLLEGE GENDER EARNINGS GAP BY OCCUPATION

Notes: The gender pay gap for college graduate is from Figure 2, part B; the O*Net characteristics are described in Table 2. Only occupations with more than 25 male and more than 25 female observations for the college, full-time sample are included in the figure. Standard errors are in parentheses under coefficients.

Source: See Figure 2.

- The figure plots output against time input for several jobs.
- High-paying jobs have steeper slopes but also bigger drops when time input falls below a threshold.
- Lower-paying flexible jobs have smaller or zero penalties.

Economic message. The framework gives an intuitive reason why workers who value flexibility sort into lower-paying positions even when they have the same formal training.

5.6 Table 3 and Figure 6: nonlinear pay among lawyers

Read:

- For JDs, the raw year-15 female earnings gap is about -0.549 log points.
- Adding log hours and log weeks cuts it to about -0.215 .
- Adding job tenure, years not employed, and years part-time reduces it further to about -0.130 .
- The coefficient on log hours at year 15 is greater than one (about 1.34 in one specification and 1.162 with fuller controls), showing convex annual pay with respect to hours.
- Figure 6 shows that both annual earnings and hourly fee rise sharply with hours; the fraction female and the fraction with children fall as hours rise.

Economic message. Law is not merely a long-hours profession. It is a profession in which *extra* hours are especially highly rewarded.

5.7 MBA and pharmacy evidence: nonlinear versus linear worlds

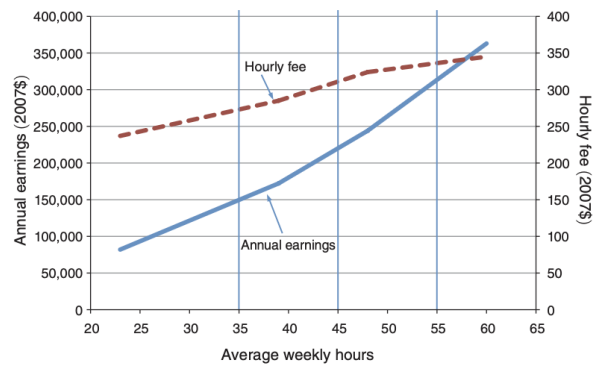
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TABLE 3—EARNINGS EQUATIONS FOR JDs: UNIVERSITY OF MICHIGAN LAW SCHOOL ALUMNI SURVEY, LONGITUDINAL SAMPLE

	Dependent variable: log (annual earnings)						log (hourly fee)	
	Year 5			Year 15			Year 15	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Female	-0.100 (0.0262)	-0.0375 (0.0240)	-0.0158 (0.0228)	-0.549 (0.0498)	-0.215 (0.0456)	-0.130 (0.0455)	-0.00345 (0.0366)	0.0302 (0.0368)
log(hours per week)		0.757 (0.0497)	0.563 (0.0521)		1.34 (0.0758)	1.162 (0.0814)	0.442 (0.0541)	0.312 (0.0589)
log(weeks per year)		0.522 (0.0771)	0.331 (0.0752)		0.846 (0.123)	0.711 (0.120)		
Years in current job			0.0423 (0.00615)		0.0251 (0.00394)			0.00886 (0.00280)
Years not employed by year t			-0.379 (0.0545)		-0.0406 (0.0274)			-0.0683 (0.0253)
Years part-time by year t			-0.244 (0.0339)		-0.0533 (0.0115)			-0.0263 (0.00854)
Time off, BA to law school			-0.0283 (0.00800)		-0.0649 (0.0145)			-0.0257 (0.0107)
<i>Dummy variables</i>								
Law school performance	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Survey year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Missing job experience	No	No	Yes	No	No	Yes	No	Yes
Missing weeks per year	No	Yes	Yes	No	Yes	Yes	No	No
Constant	11.5 (0.0161)	6.35 (0.329)	8.00 (0.340)	12.2 (0.0296)	3.70 (0.487)	4.78 (0.504)	3.94 (0.213)	4.41 (0.231)
Observations	1,449	1,449	1,448	1,299	1,299	1,298	695	694
R^2	0.0747	0.245	0.332	0.129	0.359	0.408	0.238	0.287

Notes: Regression sample includes individuals working > 9 hours per week in the given year. Law school performance includes law school GPA at graduation and whether on law review. Sample for columns 7 and 8 is restricted to those working in law firms who reported an hourly fee. Weights are the inverse of the predicted values from a probit regression on whether the survey was returned using a set of predetermined variables such as law school GPA. Standard errors are in parentheses.

Source: University of Michigan Law School Alumni Survey Research Dataset, Longitudinal Sample for individuals graduating from 1982 to 1991 who returned both the 5-year and 15-year surveys.



Job-hours feature (mean)	10 to 34 hours	35 to 44 hours	45 to 54 hours	55 hours plus
Fraction of men in bin	0.021	0.165	0.486	0.328
Fraction of women in bin	0.243	0.218	0.359	0.180
Fraction female (0.289)	0.830	0.353	0.229	0.185
With children (0.698)	0.852	0.798	0.617	0.536
<i>Work setting:</i>				
Law firm (0.487)	0.425	0.403	0.514	0.534
Corporate (0.150)	0.0755	0.135	0.184	0.148
Government (0.104)	0.0943	0.160	0.127	0.0456
Non-practicing (0.210)	0.283	0.265	0.155	0.241
<i>If in a law firm:</i>				
Partner (0.690)	0.311	0.490	0.739	0.809
Solo practice (0.0947)	0.233	0.177	0.0646	0.0660
Firm size mean (252)	188	252	249	278
Firm size median (125)	30	50	140	135
<i>Client:</i>				
Fortune 500 (0.248)	0.182	0.228	0.234	0.293
"Rich" (0.0904)	0.0241	0.0900	0.0963	0.0974

Notes: Data given here are for those working > 9 hours a week at year 15. All monetary amounts are in 2007 dollars. The category "other" accounts for the remaining fraction for work-setting. "Firm size," "Fortune 500," and "Rich" are all conditional on working in a law firm. Client variables are the (average of the) fraction of time lawyers stated they worked with such clients.

Source: University of Michigan Law School Alumni Survey Research Dataset, Longitudinal Sample for individuals graduating from 1982 to 1991 who returned both the 5-year and 15-year surveys.

FIGURE 6. HOURS, EARNINGS, AND CHARACTERISTICS OF OCCUPATIONS FIFTEEN YEARS AFTER THE JD

- For Chicago Booth MBAs, men and women start essentially together, but the gap reaches about 30 log points after five years and roughly 60 log points by 10–16 years after the MBA.
- Goldin reports that about 84% of the late-career MBA gap is explained by prior training (24%), career interruptions and job experience (30%), and weekly hours (30%).
- For pharmacists, part-time work is common and there is little part-time penalty.
- Structural change—large chains, standardization, information systems, and easier handoffs across workers—made pharmacists much better substitutes for one another.

Economic message. MBA and law look like the convex-pay model. Pharmacy looks like the linear-pay alternative.

6 Short summary

- The paper asks what explains the remaining gender pay gap once women have largely caught up in education and training.
- The main answer is that the residual gap is strongly linked to job structure, especially whether earnings are nonlinear in hours worked.
- Women and men begin their careers with much more similar earnings than they have later on.
- Most of the contemporary gap is within occupations rather than between occupations.
- Business and law exhibit large residual gaps and strong convexity of pay with respect to hours.
- Technology, science, and especially pharmacy look more egalitarian because workers are more substitutable and flexibility is less costly.
- The broader lesson is that closing the last part of the gender gap requires redesigning jobs so that long or particular hours are not disproportionately rewarded.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the remaining gender pay gap in high-skill labor markets is tightly linked to the organization of work. The modern gap opens up over the life cycle, differs sharply across occupations, and is largest in jobs that impose heavy penalties on reduced hours or temporal flexibility.

7.2 Economic intuition

When workers are imperfect substitutes and information, clients, or responsibility cannot be handed off cheaply, firms reward long and continuous hours more than proportionally. That creates a large

price for flexibility. Because women, especially mothers, more often adjust hours and continuity, they bear a disproportionate earnings penalty in such occupations.

7.3 Takeaway

The paper's message is not simply that gender inequality remains. It is that the *form* of remaining inequality has changed. The “last chapter” requires a labor market in which workers can substitute for each other more seamlessly and in which flexible time no longer comes at such a steep wage cost.